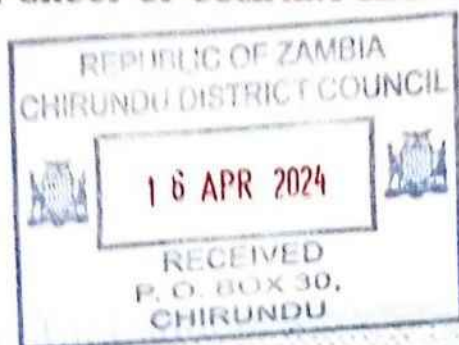


**FINAL ENVIRONMENTAL PROJECT BRIEF
FOR THE PROPOSED CONSTRUCTION OF A FEED STORAGE
WAREHOUSE IN CHIEF SIKONGO'S AREA OF CHIRUNDU DISTRICT BY
YONG WANG INVESTMENT GROUP OF COMPANY LIMITED.**



April, 2024

Prepared by Elizkoffi Consult Limited

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project

EXECUTIVE SUMMARY

Background

Yong Wang Investment Group of Company Limited is a Zambian company incorporated in 2020. The company is in the process of setting up fish cages on the Kafue River after receiving approval from both the Zambia Environmental Management Agency and the Water Resources Management Authority. The company intends to construct a warehouse for storage of fish feed, nets and housing of a cold room for ice making.

Project Description

Yong Wang Investment Group of Company Limited intends to put up the following on a 2hactares piece of land:

- A 29.7m by 11.7m by 6.0m warehouse for storage of fish feed and fish nets. The warehouse will also house a cold room that will be used for making ice blocks;
- A small conveyor belt that will be used for ferrying feed to the feeding area;
- Silt traps that will prevent siltation;
- Culverts that will allow the continuous natural flow of storm water and;
- Living quarters for workers.

Project Location

The proposed site is located off Chiawa (RD 491) road near Michael Sata Bridge (locally known as Chiyaba Bridge). The site can be accessed using Chiyaba road and it is 11 km from Chirundu border and about 152 Km from Lusaka CBD. The main landmark is Chiyaba Bridge. The site lies on GPS coordinates 15°56'38.4176"S 28°52'23.1546"E.

Project background and rationale

Yong Wang Investment Group of Company Limited is a Zambian company incorporated in 2020. The company holds an EIA approval (Decision letter) to develop fish cages on the Kafue River to produce around 70tons of fish per annum. The company intends to set up a small warehouse about 60 meters from the Kafue river to store fish feeds and nets.

Project Objectives

The main objective is to construct a warehouse for fish feed storage. The warehouse will also house a cold.

Project Investment

Approximately, K700,000 will be invested in this project.

Alternatives Considered

The alternatives considered included;

- I. Project Location

project

Options considered included developing the project at the proposed site or finding another site. The proponent considered another side but forewent it due to economic reasons. Constructing at the proposed site was the chosen option.

II. No Project Option

This option was not considered given the economic benefits that would arise from implementation of the proposed project.

III. Energy source: Connection to Zesco was the chosen option

IV. Water Source: for operation phase, drilling of a borehole was the chosen option

V. Sewage Management: use of septic tank was the chosen option,

VI. Storm water management: use of culverts was the chosen option.

Legal Framework

The project has identified a number of legal and institutional frameworks to comply with. They have been fully explained in the report and they include the following;

- The Environmental Management Act (EMA) No.12 of 2011.
- The Environment Management Act (Licensing) Regulations (SI No. 112 of 2013) *Part II*
- Water Resources Management Act of 2011
- The Fisheries Act No. 22 of 2011
- Food and Drug Act (Cap 303)
- The Local Government Act No. 2 of 2019
- The Workers Compensation Act No. 10 of 1999
- The Occupational Health and Safety Act, 2010
- National Council for Construction Act No.10 of 2020
- Zambia Wildlife Act No. 14 of 2015
- International Agreements and Conventions

Institutional Framework

1. The Zambia Environmental Management Agency
2. Ministry Local Government – Chirundu Town Council
3. Water Resources Management Authority
4. The Department of National Parks and Wildlife

This Environmental Project brief which includes an Environmental and Social Management Plan (ESMP) has identified the following among others:

Negative Impacts

1. Dust and noise generation
2. Disturbance of natural drainages
3. Siltation as a result of construction activities
4. Animal – human conflict

project

5. Generation of solid and liquid waste,
6. Disturbance of vegetation and biodiversity,
7. Surface water contamination
8. Loss of aesthetics

The proponent will implement the following measures to mitigate the negative impacts;

1. Limiting construction activities to day time only (06-18hours)
2. Installation of culverts to ensure natural flow of water
3. Installation of silt traps to prevent siltation
4. Limiting vegetation clearance only to area of interest
5. Implementation of proper waste management measures to prevent indiscriminate waste disposal
6. Ensuring the structures blend with the areas aesthetics

CONCLUSION AND ENVIRONMENTAL COMMITMENTS

Yong Wang Investment Group of Company Limited plans to develop and implement an effective ESMP to address all the projected negative impacts, ensure effective parameter monitoring and proper alignment to subsequent audits.

The ESMP summarizes mitigation measures to be taken with respect to the different environmental issues involved and the responsibilities of key stakeholders at the various project phases.

Key parameters to monitor during operation of the project will include:

- Solid waste storage, collection and disposal;
- Surface water contamination;
- Compliance with public health standards and requirements and;
- Human – Human conflicts
- Eutrophication
- Drainages in the project area



Director

Yong Wang Investment Group of Company Limited

project

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project

ABBREVIATIONS AND ACRONYMS

AIDS	Acquired Immune Deficiency Syndrome
°	Degrees
°C	Degrees Celsius
DNPW	Department of National Parks and Wildlife
EA	Environmental Audit
EIA	Environmental Impact Assessment
EMA	Environmental Management Act
EPB	Environmental Project Brief
GMA	Game Management Area
GPS	Global Positioning System
GRZ	Government of the Republic of Zambia
Ha	Hectare
HIV	Human Immunodeficiency Virus
IUCN	International Union for Conservation of Nature
Km	Kilometre
Km ²	Square Kilometre
'	Minute
Mm	Millimetre
%	Percentage
PA	Protected Area
UNESCO	United Nations Educational, Scientific and Cultural Organisation
USD	United States Dollar
UTM	Universal Traverse Mercator
ZEMA	Zambia Environmental Management Agency

1 INTRODUCTION

1.1 Project Background

Yong Wang Investment Group of Company Limited is a Zambian company incorporated in 2020. The company is in the process of setting up fish cages on Kafue River after receiving approval from both the Zambia Environmental Management Agency and the Water Resources Management Authority. The company intends to construct a warehouse for storage of fish feed, nets and housing of a cold room for ice making.

1.2 Project Objective

The main objective is to construct a warehouse for fish feed storage and cold room housing for an aquaculture project.

1.3 Objectives of the Environment Project Brief

The objective of this EPB is to determine, define and evaluate the foreseeable immediate and long-term impacts of implementation of the aquaculture project. Specific objectives include:

- To identify and evaluate the environmental impacts that will be caused by the project activities.
- To identify and describe procedures and activities that will mitigate adverse impacts and enhance any beneficial impacts of the project.
- Recommend mitigation measures for the adverse impacts of the aquaculture project on the biodiversity of the site
- Develop an environmental and social management plan (ESMP) for the project undertaking.

1.4 Project Developer, Physical Address and Contact Person

Particulars of Shareholders and Directors

Table 1 Particulars of Directors

NAME	NATIONALITY	PASSPORT No.	POSITION	PERCENTAGE (%) SHARE
Lin Denghui	Chinese	EB5825493	Director	51
Zhuang Mei	Chinese	EB9123193	Director	49

Developers details

Yong Wang Investment Group of Company Limited
Plot No. LUS14031
Off Katimamulilo road,
Olympia Park Extension
Lusaka.

project

Tel: +260 960 655 555
414173045@qq.com

Contact Person Details.

Contact Person: Mr Akinbayode Ezekiel
Designation: Representative
Plot No. 2, Airport road
Lusaka.
Tel: +260 975915065
ezeakin@hotmail.com

1.5 Track Record

Yong Wang Investment Group of Company Limited has been in business since 2020 in Zambia and this is the first project they are embarking on in aquaculture. However, the directors have been in the aquaculture business for over 25 years back in China.

1.6 Project Location

The proposed site is located off Chiawa (RD 491) road near Michael Sata Bridge (locally known as Chiyaba Bridge). The site can be accessed using Chiyaba road and it is 11 km from Chirundu border and about 152 Km from Lusaka CBD. The main landmark is Chiyaba Bridge. The site lies on GPS coordinates 15°56'38.4176"S 28°52'23.1546"E.

1.7 Project Description

Yong Wang Investment Group of Company Limited intends to put up the following on a 2 hectares piece of land:

- A 29.7m by 11.7m by 6.0m warehouse for storage of fish feed and nets. The warehouse will also house a cold room that will be used for making ice blocks;
- A small conveyor belt that will be used for ferrying feed to the feeding area;
- Silt traps that will prevent siltation;
- Culverts that will allow the continuous natural flow of storm water and;
- Living quarters for workers

1.8 Total Project cost/ investment

The proposed development will be completed at an estimated cost of K700,000.

1.9 Proposed project implementation date

Implementation will commence immediately after The Zambia Environmental Management Agency (ZEMA) approves the EPB report.

2 LEGAL AND POLICY FRAMEWORK

This Environmental Project Brief was prepared in accordance with the legal framework on Environmental Management enshrined in the Environmental

Management (Amendment) Act No. 8 of 2024 as read together with the EMA No. 12 of 2011 and its subsidiary legislation, the Environmental Impact Assessment Regulations S.I. No. 28 of 1997. Section 3 (1) of the EIA Regulations which states that, “a developer shall not implement a project for which a project brief or environmental impact statement is required under these Regulations, unless the project brief or the environmental impact statement has been concluded in accordance with these regulations and the Zambia Environmental Management Agency (ZEMA) has issued a decision letter.”

2.1 Policy, legal and institutional framework relevant to the Project

The Policy, legislation and institutional framework requirements associated with the development of the site include among others the following: -

2.2 Environmental Management (Amendment) Act No. 8 of 2024 as read together with the EMA No. 12 of 2011.

The EMA Act is the principal Act governing and regulating environmental issues in Zambia and provides specific regulation for discharge, collection, storage, transportation and disposal of gaseous, liquid and solid waste. Its main functions include the protection of the environment and control of pollution. In particular, it provides for the health and welfare of people, animals, plants and the environment.

Relevance; section 29 subsection (1) of the Act states that “a person shall not undertake any project that may have an effect on the environment without the written approval of the agency, and except in accordance with any conditions imposed in that approval. Section 30 subsection (2) of the Act states that “the categories of projects that are considered to have an effect on the environment for the purpose of subsection 1 of twenty-nine (29) and are required to conduct an environmental impact assessment. Therefore, being a development that will be located on an Environmentally sensitive area, an Impact Assessment will be required.

Compliance there of: This project report is being prepared in compliance with the above legal provisions.

2.3 Zambia Wildlife Act No. 14 of 2015

Enacted in 2015, the Act provides for the conservation and management of ecosystems to preserve them from the impacts of modern man. The Act regulates the types and extent of activities that may be permitted in National Parks and Game Management Areas, also providing for the sustainable use of wildlife and the effective management of wildlife habitat in Game Management Areas, thereby enhancing the benefits of Game Management Areas to local communities and wildlife.

project

Relevance: The Act is very applicable owing to the site being located just outside the GMA and about 70m from the Kafue River, which is known to have presence of wildlife.

Compliance there of: The developer shall ensure that the provisions of the Act are adhered to and implemented.

2.4 The Fisheries Act No. 22 of 2011

Provides for regulation of aquaculture development planning and aquaculture licensing, control of fish movements and introductions. The introduction of any species of fish or the importation of any live fish without the written permission of the Director of Fisheries is prohibited. There is no provision for traceability of authorised introductions such as control of subsequent translocations within the country because the 'certificate of origin' is only used for dead fish and not for control of live fish movements across district boundaries. Equally absent are sanitary guidelines on importation of live fish specimens. Similarly, there is no control for the species of aquatic life that may be introduced into any aquaculture facility or provisions for regulating location, design and materials used in the construction of the aquaculture facility and to the equipment that used therein. The Fisheries Act No. 22 of 2011 is silent on control and monitoring of water quality for aquaculture purposes. However, it is very clear on measures to safeguard the property rights in aquaculture by restricting unauthorised interferences and obstructions to an aquaculture facility. It also prohibits the use of fish growth enhancing hormones.

Relevance: the warehouse being developed is an auxiliary facility for an aquaculture project.

Compliance there of: the proponent will ensure that no chemicals such as growth enhancing hormones are not used.

2.5 The Public Health (Infected Areas) (Corona Virus Disease 2019) Regulations, 2020, Statutory instruments 21 and 22.

This provides for procedures to control and prevent the spread of Corona Virus Disease 2019 (COVID-19) in the Zambia providing guidelines providing for among others; sanitation and hygiene in public premises, prohibition or restriction on trading or vending in unsanitary conditions, compulsory notification of cases or suspected cases and the restriction of the number of attendees of a public ceremony or gathering within an infected area.

Relevance: The project will have gatherings during constructions and operation phases and also storage of fish feed.

Compliance there of: The developer shall ensure strict adherence to and implementation of the provisions of the regulations.

2.6 National Heritage Conservation Act (CAP 173)

Enacted in 1989, the Act provides for the Conservation of Ancient, Cultural and Natural Heritage, Relics and Objects of Aesthetic, Historical, Pre-historical, Archaeological or Scientific interest. It established the National Heritage Conservation Commission (NHCC) and sets out its functions.

Relevance: The Act is relevant as the project will take place on the Kafue River and will be having divers that will be maintaining the project cages.

Compliance there of: The project will inform NHCC of any heritage or suspected heritage findings on the site.

2.7 Lands Act of 1995 and Land Acquisition Act of 1995

The Lands Act is administered by the Department of Lands for the alienation of land under statutory leaseholds. The Lands Act categorises land as State, Local Authority and Traditional Land, providing for the continuation of leasehold tenure; to provide for the continued vesting and alienation of land by the president; to provide for the statutory recognition and continuation of customary tenure; to provide for the conversion of customary tenure into leasehold tenure; to establish a land development fund and a Lands tribunal. The proposed developments fall under Traditional (customary) Land.

Relevance: The project site is under traditional land hence consent from the traditional leadership will need to be obtained.

Compliance there of: Consent from the Chief was obtained.

2.8 Local Government Act No. 2 of 2019

An Act to provide for an integrated local government system; give effect to the decentralisation of functions, responsibilities and services at all levels of local government. ensure democratic participation in, and control of, decision making by the people at the local level; revise the functions of local authorities; provide for the review of tariffs, charges and fees within the area of a local authority; provide for the proceedings of the council and committees; provide for the role of traditional leadership in democratic governance; repeal and replace the Local Government Act, 1991; and provide for matters connected with, or incidental to, the foregoing

Relevance: The project will abide by the By-Laws established by the Chirundu Town Council.

Compliance there of: Acquisition of Local Authority permits and other environmental regulations as stipulated therein.

2.9 The Employment Code Act No. 3 of 2019

Enacted in 2019, The Act provides for regulating the employment of persons; prohibiting discrimination at an undertaking; constituting the Skills and Labour Advisory Committees and providing for their functions. It provides for the engagement of persons on contracts of employment, providing for the form and enforcement of the contracts, entitlements and other benefits of

project

employment, including the protection of wages of employees. The Act regulates the employment of children and young persons in an undertaking.

Relevance: Being a project that will employ people to work either as permanent or as casual workers during the construction and operation of the project.

Compliance thereof: The developer will ensure that the interests of the people who will be employed throughout the project are safeguarded and that employees are remunerated in accordance with the provision of the Act.

2.10 Water Resources Management Act No. 21 Of 2011.

The water Resources Management Act No. 21 of 2011 replaces the Water Act, 1949. The Act establishes the Water Resources Management Authority and define its functions and powers as well as provide for the management, development, conservation, protection and preservation of the water resource and its ecosystems. This Act also provides for the equitable, reasonable and sustainable utilisation of the water resource as well as ensure the right to draw or take water for domestic and non-commercial purposes, and that the poor and vulnerable members of the society have an adequate and sustainable source of water free from any charges and create an enabling environment for adaptation to climate change. Sections 3 and 4 of Part I of this Act prohibits the absolute ownership of natural water bodies, while Part V provides guidance on pollution of water bodies in Zambia.

Relevance: being a project that will be located near the Kafue river and will require drilling of a borehole

Compliance there of: The project proponent shall ensure strict adherence to the provision of this Act.

2.11 Solid Waste Management Act No. 20 of 2018

The Act provides for an array of matters to do with the sustainable regulation and management of solid waste including; general and self-service solid waste services; the incorporation of solid waste management companies and defining their statutory functions; the licensing and functions of solid waste service providers, operators and self -service solid waste providers and providing or their functions; the regulation, operation, maintenance and construction of landfills and other disposal facilities; the setting and approval of tariffs for management of solid waste and provision of solid waste services; and matters connected with, or incidental to, the foregoing.

Relevance: The Act is relevant as the project that will generate solid waste throughout all the phases of development and operations.

Compliance there of: The project will work with the local authority and ensure that the provisions of this act are adhered to.

project

2.12 The National HIV/AIDS/STI/TB Act of 2002 Section 4

The Act provides for the establishment of the National HIV/AIDS/STI/TB council and provides for the support to the development, monitoring and evaluation of multi-sectorial and economic impacts of HIV, STI, AIDS, and TB.

Relevance: Since many will be empowered it might result in workers indulging in casual sexual activities which might lead to the spread of the HIV/AIDS and other STIs.

Compliance there of: Awareness programs will be arranged for all the workers as well as the distribution of condoms to the employees.

2.13 The Standards Act No. 4 of 2017

An Act to continue the existence of the Zambia Bureau of Standards and re-define its powers and functions; provide for standardization and quality assurance of products and services through the setting of national standards and provision of conformity assessment services for products and services; repeal the Standards Act, 1994; and provide for matters connected with, or incidental to, the foregoing.

Relevance: The project will involve purchasing of building materials and construction of a warehouse and cold room equipment.

Compliance there of: The construction materials will meet the requirements of the Zambian Bureau of Standards.

2.14 Roads and Traffic Control Act No. 11 of 2002

The Act establishes the Road Transport and Safety Agency which was mandated to provide for a system of road safety and traffic management.

Relevance: The contractor will use different vehicles on the road during construction. Further, the proponent trucks will be accessing the proposed site for deliveries.

Compliance: Proponent will ensure the use of the road in accordance with these regulations by operating before and after traffic busy-hours and using proper road signage.

2.15 Institutional Framework

To ensure the successful implementation of the Project Brief and the safeguards contained herein, the following institutions have been identified as critical to project coordination, monitoring and evaluation.

2.16 The Zambia Environmental Management Agency

The Zambia Environmental Management Agency (ZEMA) is the umbrella environmental institution in Zambia and the main lead agency on matters pertaining to environmental impact assessments (EIA). It is empowered by the Environmental Management (Amendment) Act No. 8 of 2024 as read together with the EMA No.12 of 2011 to identify projects, plans and policies for which

an EIA is necessary. The general functions of ZEMA are to ensure the sustainable management of natural resources, the protection of the environment and the control of pollution. However, more specifically, ZEMA serves *inter alia* to:

Co-ordinate the implementation of activities of all government ministries, appropriate authorities and conservancy authorities in matters relating to the environment.

2.17 Department of National Parks and Wildlife.

The Department of National Parks and Wildlife (DNPW) is governed by law through the Zambia Wildlife Act No. 14 of 2015 and under the guidance of the Ministry of Tourism and Arts to protect and conserve Zambia's wildlife estate, improving the quality of life among communities and reducing the decline in wildlife resource, however, maintaining sustainable bio-diversity in National Parks and GMAs to levels which will secure sustainable flow of benefits from the resources.

2.18 Local Authorities – Chirundu Town Council

Local Authorities are concerned with the health of local community members and conservation of natural resources within the confines of their administrative influence. The local authorities are responsible for enforcement of the provisions of the Local Government and Public Health Acts. The councils play a key role in revenue collection, service provision, local environment management and advocacy.

2.19 The Department of Fisheries – Ministry of Livestock and Fisheries

The department of fisheries is guided by the Fisheries Act which provides for the development of commercial fishing, enforcement of fishing regulations and laws such as fish ban. The Act also cites and identifies aquaculture as a viable means of developing fisheries in Zambia and promotes environmental stewardship and application of international codes on good aquaculture practices in line with the SADC Protocol on fisheries.

2.20 Water resources management Authority (WARMA)

The Water Resources Management Authority (WARMA) was established by the Water Resources Management Act to ensure the management, development, conservation, protection and preservation of the water resource and its ecosystems as well as equitable and sustainable utilization of the water resource. Permits will thus need to be sought from WARMA and the company will abide by any condition, limitation, restriction or prohibition that WARMA may impose for the sustainable utilization of the water in the rivers.

2.21 International Conventions and protocols

Zambia - being a signatory and party to more than thirty (30) International Conventions and Protocols among which the most relevant include; the Convention on Biological Diversity of 1992 and ratified in 1993, the

convention on International Trade in Endangered Species of the wild fauna and Flora (CITES) of 1973 and enforced in 1975, the Convention concerning the Protection of World Cultural and National Heritage (ratified in 1972), the United Nations Framework Convention on Climate Change, the African Convention on the Conservation of Nature and Natural Resources and The Ramsar Convention on Wetlands of International Importance. The table below explains the various international conventions and treaties relevant to the project.

Relevance: The Conventions are relevant to the project as the activities embarked by the developer may have an impact on the resources identified.

Compliance there of: Minimizing or avoiding significant adverse effects where appropriate on the identified natural resources.

Table 2 International Environment Conventions to which Zambia is Signatory

Convention	Description	Relevance to the Project
African Convention on the Conservation of Nature and Natural Resources. (Algiers 1968)	The Convention encourages conservation, utilisation and development of soil, water, flora and fauna for the present and future welfare of mankind from an economic, nutritional, cultural and aesthetic point of view	The project will be located near the Kafue that will need to be sustainably utilization
Convention on Biological Diversity. (1992)	Encourages the conservation of biological diversity, the sustainable use of its components and the fair and equitable sharing of benefits arising out of the utilisation of genetic resources.	The project will be near the Kafue River
Convention on International Trade in Endangered Species- CITES (1973)	Convention aims at ensuring that the internal trade in specimen of wild animals and plants does not threaten their survival.	The project is located near the GMA and also an area rich in water fauna and the project operators might have access to endangered species.
Convention concerning the Protection of World Cultural and National Heritage (1972)	The convention aims at ensuring the identification, protection, conservation, presentation and transmission to future generations of the cultural and natural heritage	Project activities such as construction might expose or reveal artefacts that of national value

project

United Nations Framework Convention on Climate Change – UNFCCC (1997)	The goal of the convention is to stabilise Green House Gas (GHG) concentrations at a level that would prevent anthropogenic interference with the climate system.	Construction activities and certain practices might contribute to GHG emissions. Also the cold room might use refrigerants that may be GHG
Basel Convention on the control of Transboundary movements of hazardous wastes and their disposal (2005)	The convention aims to control import and export of hazardous waste across regions and international boundaries.	Hazardous waste might be generated from spills and leakages
RAMSAR Convention on Wetlands of International importance especially as Waterfowl Habitat. Ratified: 1975	The convention aims at protecting wetlands through sustainable utilisation of them.	The project site is near the Kafue River

3 DESCRIPTION OF THE PROJECT

3.1 Location

The proposed site is located off Chiawa (RD 491) road near Michael Sata Bridge (locally known as Chiyaba Bridge). The site can be accessed using Chiyaba road and it is 11 km from Chirundu border and about 152 Km from Lusaka CBD. The main landmark is Chiyaba Bridge. The site lies on GPS coordinates 15°56'38.4176"S 28°52'23.1546"E.

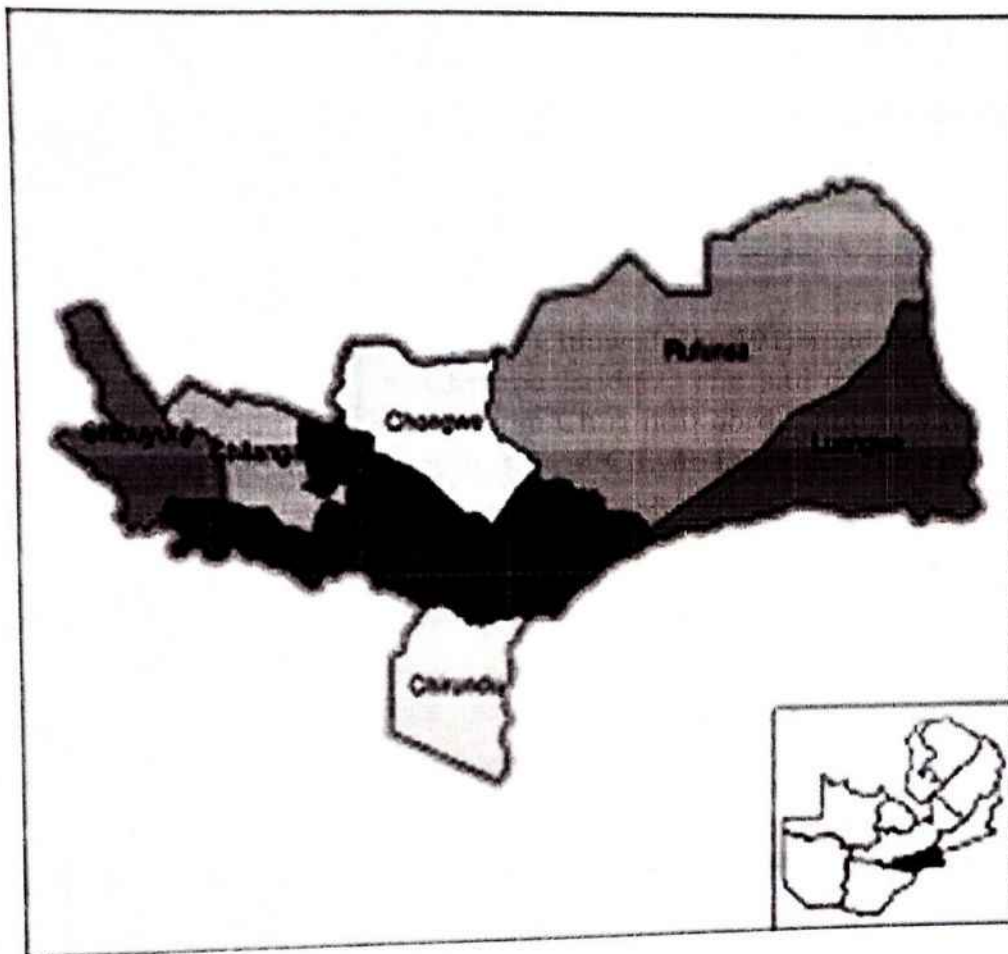


Figure 1: Location Map for Chirundu

Table 3: Co-ordinates of the Proposed site

CO-ORDINATES		
CORNERS	S	E
A	15°56'38.4176"	28°52'23.1546"
B	15°56'36.4915"	28°52'31.4079"
C	15°56'37.6585"	28°52'31.8894"
D	15°56'38.9341"	28°52'31.0944"
E	15°56'40.3666"	28°52'23.4753"



Figure 2: Location Map of the proposed site

project

3.2 Nature of the Project

Yong Wang Investment Group of Company Limited intends to put up the following on a 2 hectares piece of land:

- A 29.7m by 11.7m by 6.0m warehouse for storage of fish feed and fish nets for fish cages. The warehouse will also house a cold room that will be used for making ice blocks;
- A small conveyor belt that will be used to ferry feed to the feeding area;
- Silt traps that will prevent siltation;
- Culverts that will allow the continuous natural flow of storm water and;
- A Workers quarters.

Raw Materials

Table 4 below, shows the typical raw materials to be utilized for construction activities and their possible sources.

Table 4: Raw Materials and their possible sources

Raw Materials	Sources
Cement	Chirundu
Building blocks	Licensed supplier – Kafue
Roofing sheets	Lusaka
Conveyor belt	Lusaka
Steel bars	Lusaka
Fish feed	Lusaka
Cold room equipment	China

Products and by products

Once implemented, the project will have a storage facility and an ice making cold room. No by products will be produced from the project as it is a construction project.

Production capacity

The warehouse will measure 11.7 by 29.7 by 6 meters giving a total volume of 2,084 cubic meters. However, only about 70% of this will be utilized for storage and cold room as room will be left for proper stacking height and spacing.

Schedule and life time of the project

Construction activities will commence as soon as ZEMA gives approval of the Environmental Project Brief Study. The project lifespan is expected to be 20 years.

3.3 Main activities

The main activities of the project will comprise three phrases namely, preparation, the Construction and the Operational phases.

Preparation Phase

The phase basically involved site selection and suitability studies. Other activities will include procurement of materials, obtaining the necessary permits and engaging workers and contractors. Other activities will include land clearing and leveling.

Construction Phase

The following are among the main activities that will be undertaken during the construction phase: -

1. Metal fabrication of the warehouse pillars
2. Trenching and foundation works
3. Concrete works and backfilling
4. Actual bricklaying and roofing
5. Installations of culverts
6. Setting up of silt traps
7. Installation of cold room equipment
8. Installation of a conveyor belt

3.4 Operational Phase

This stage will involve the running and managing of the facility as per the laid down rules and procedures. The main activities that will be undertaken in during this phase will include among others: -

1. Loading and offloading of feed from the warehouse
2. Movement of new and old fish nets in and out of the warehouse
3. Making of ice blocks in the cold room during fish harvest period
4. Maintenance works for the warehouse and auxiliary facilities
5. Movement of trucks in and out of the project area
6. Continuous monitoring of environmental issues and implementation of the Environmental and social management plan

4 PROJECT ALTERNATIVES

An array of alternatives has been analyzed with the aim of identifying the most suitable options for implementation either as a single or a combination of one, two or all available options for the management of the project.

project

Identification of Alternatives

Options such as project need, project location and project design were considered

4.1 Project Need

The aquaculture project that will be implemented by the company will require procurement of huge volumes of fish feed and maintenance of fish nets that will be damaged through tear and wear and predators. Chirundu has no sources of fish hence the product will need to be procured from Lusaka which entails high transportation costs and intermittent availability. The construction of the warehouse will ensure continuous supply of feed, lower transportation cost and reduced inconveniences. Further, the operation of the warehouse will help provide jobs to the local community members.

4.2 Site Alternatives

Option A – Develop the Project at the proposed site.

This will mean a shorter distance to the aquaculture project site hence reduction of Environmental impacts. Further, the proponent has already leased the land for the project development and consent was granted by the Chief.

Option B – Develop the Project at another site

Another option was to develop the project at another site. This meant looking for another site either on the Kafue river or Lake Kariba.

Option A was the chosen alternative as the area is wide enough and the shore area has accommodation which the proponent could rent for easy access to the river.

4.3 Source of Energy

Two options were considered. Connection to the national grid (through Zesco) or use of either diesel-powered generator or solar energy. The area Zesco hence connection to the national grid was the chosen option.

4.4 Sewage Management

Two options were considered. Connection to a sewerage utility company or use of septic tanks. The site is on lease and previously had pit latrines. Construction of a septic tank at over 50meters from the river was the chosen option.

4.5 Source of water

Two options were considered. Drilling of a borehole or abstraction of water from the Kafue River. The first option was the chosen option.

4.6 Storm water management

The site has two small gullies that carry storm water during the rain seasons. Two options were considered for storm water management. That is,

project

maintenance of the old gullies or use of culverts. The old gullies were known to cause minor flooding hence culverts will be used to ensure non disturbance of the natural flow of water.

4.7 Silt management

Silt can be a problem during construction phase and the rain season. Two options were considered for management of silt. Concrete lining the whole yard or use of silt traps. Use of silt trap (wire gauze with stones) was the chosen option.

project

5 DESCRIPTION OF THE BASELINE ENVIRONMENT

5.1 Climate

The climate at the site is typical of the Southern province region with dry, wet and hot seasons during the different times in a year. The hot and wet rainy season lasts between November to April. The rains are usually accompanied by some thunderstorms and lightening. The hot and dry season lasts between August and October. The daily maximum and minimum average temperatures of 25°C and 14.2°C occur in October and July respectively.

5.2 Temperature

The mean monthly temperatures range from a minimum of 14.2°C to a maximum of 31°C. The average temperature is 25°C.

5.3 Wind

The prevailing winds at the site are from the east to the west direction. The Annual Mean Wind Speed is estimated at 13kph.

5.4 Annual Rainfall

Rainfall occurs mainly during October to May. The mean annual rainfall for the region is 129mm.

5.5 Geology and Soils

The geology of the Lower Zambezi area comprises a varied suite of rocks of various origins and ages. The youngest rocks are of colluvial materials found in the valley floor. Close to the valley floor are red sandstone, basalt and sandstone all belonging to the Karroo system. A variety of rocks, such as biotite schists, quartzite, meta-quartzite, granite gneisses calc-silicate and carbon rocks, distributed widely in the region, were produced by ancient volcanic activities and metamorphism in the region (Barr, 1971). The proposed bridge site is underlain by alluvial soils extending to a depth of at least 5 meters. The nature of the alluvium on both sides of the bank is characterised by silty sands, silty clays, gravelly silts.

5.6 Flora

The vegetation observed around the study area are basically of five distinct woodland types namely Mopane, Munga, Riparian and dry deciduous woodlands. Of significance is a big baobab (*Adansonia digitata*) standing on the right-hand side of the road near the project that is of socio-cultural value to the area.

5.7 Fauna

During site visit no large fauna was observed on shore other than some birds, and lizards (*lacertilian*). There are no endangered bird species at the site.

Some common birds in the area include: common bulbul (*Pycnonotus barbatus*), zitting cisticola (*Cisticola juncidis*), marsh warbler (*Acrocephalus*

palustris), streaky seedeater (*Serinus striolatus*). However, the site area is known to have a lot of crocodiles and plenty of fishes.



Figure 3: cages to protect locals from crocodile attacks during laundry or dish washing

5.8 Air Quality and Noise

The air quality in the project area is generally good, except occasionally in vicinity and at the time of mechanical or vehicular activity. Around Chirundu township the air is heavily affected by emissions from trucks that are hauling goods using the road connecting Zambia and Zimbabwe. After Chirundu going towards Chiyaba the air quality tends to be generally of good quality. Similarly, the noise levels are very low as it is very far from the CBD and void of any industrial activities.

Table 5: Air quality and noise analysis results

Site	SO ₂ Level results (µg/m ³)			NO _x Level results (µg/m ³)			CO (mg/m ³)			Noise (dB)(IFC Limit=85dB)		
	1 st	2 nd	Avg	1 st	2 nd	Avg	1 st	2 nd	Avg	1 st	2 nd	Avg
S1	Trace amount	Trace amounts	Trace amounts	ND	ND	ND	0.12	0.10	0.11	43	45	44
S2	Trace amounts	Trace amounts	Trace amounts	ND	ND	ND	0.02	0.04	0.03	46	48	47
S3	ND	ND	ND	ND	ND	ND	0.11	0.11	0.11	40	42	41

project

5.9 Description of the Social Economic Environment

The site is located in Chirundu district of Southern Province, the estimated population is over 78000. The current major activities include: Farming activities, cattle ranching, trading activities along the main road, and by virtue of the District being a border town truck parking and accommodation services are the highly pronounced businesses in the district with the Zambezi and Kafue Rivers being the spectacles of tourism in the area. Other social economic aspects include formal employment in various border agencies while most of them being in government schools, clinics and other institutions.

5.10 Water Resources

The proposed project site is near the Kafue river opposite the Chiyaba Bridge. The water flow rates are in the ranges of 9-10m/s with a depth of over 15m. The river is used as a source of water for many domestic uses as well as drinking water for animals. The river near the project site is used for tourism as boat cruises are common in the area. This is due to a docking area being near the project site at the Gwabi lodge. About 7km downstream is the Kafue-Zambezi River confluence.

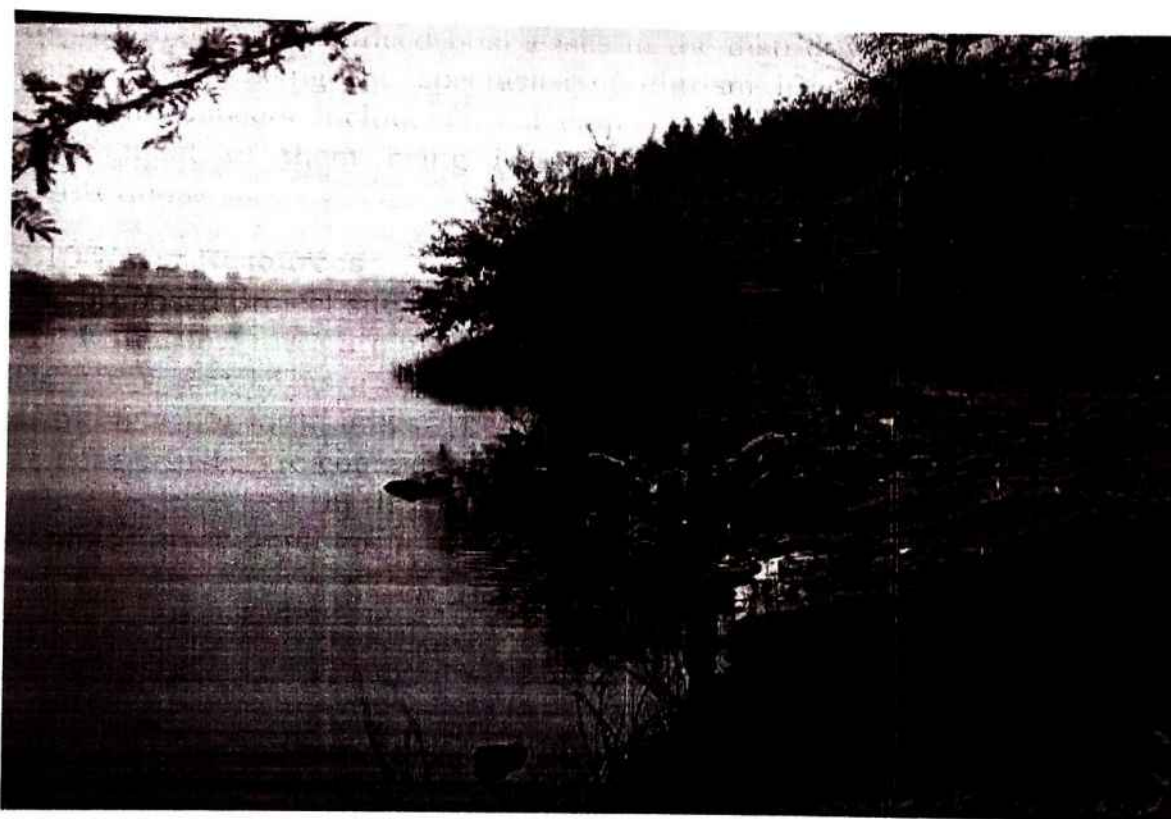


Figure 4: other water users about 40metres upstream

5.11 Water Quality

The water quality within the project site can be interpreted as good, as there are currently no significant sources of pollution along the proposed Project area. Source of pollution in the project area include boats from the Gwabi

project

Lodge which is about less than a kilometer away from the proposed project site and the Kafue Gorge Lower power project. The flowrates are high around 9-10m/s hence this helps in carrying pollutants away from the proposed project area. A sample of the water at the proposed site (Kafue River) was taken and taken for analysis. Figure 4 below gives the water quality results .



SCHOOL OF ENGINEERING
CIVIL ENGINEERING DEPARTMENT
ENVIRONMENTAL ENGINEERING LABORATORY

P.O Box 32379, Lusaka

PHYSICAL/CHEMICAL AND BACTERIOLOGICAL EXAMINATION OF WATER

Attn : Yong Wang Group of Companies

Sampled by : Client

Receipt date : 04.09.2023

Report date : 09.09.2023

Parameter	Laboratory Results		
	Water Sample	ZEMA Standards	ZABS Standards
pH	7.15	6.0-9.0	6.5-8.0
Conductivity ($\mu\text{S/cm}$)	252	4,300	1,500
Total suspended solids (mg/l)	<1.0	100	-
Total Dissolved Solids (mg/l)	126	3,000	1,000
Turbidity (NTU)	1.15	15	5.0
Biochemical Oxygen Demand (as mg O_2/l)	0.8	50	-
Chemical Oxygen Demand (as mg O_2/l)	<3.2	90	-
Sulphates (mg/l)	<0.0001	1,500	400
Nitrates (as $\text{NO}_3\text{-Nmg/l}$)	<0.0001	50	10.0
Bacteriological Results			
Feecal coliforms (#/100ml)	75	5,000	0

Tests carried out in conformity with " Standard Methods for the Examination of water and Wastewater APHA, 1998".

Tested by: D. Mkandawire
Chief Engineer

Checked & Approved by: D. Mkandawire
Chief Engineer



Figure 5: water quality results at the proposed site

5.12 Administration

Like the rest of the country, there are three administrative systems in the District and these are: Central Government, Local Government (Authority) and Traditional Systems. Central Government Administration is composed of all government departments and is headed by the District Commissioner (DC) appointed by the Republican President. The DC supervises and coordinates

project

government activities in the district. He also harmonizes the activities of the Local Authority with those of Central Government in the Districts. Heads of Departments in respective government departments in the Districts report directly to their provincial heads on technical matters and to the DC on administrative matters. Local Government Administration system is based on the political delineation of the District.

The constituency in the affected area has a duly elected member of parliament who represents the residents in the National Assembly. The Member of Parliament has a mandate covering a period of five years. Councillors are also elected to represent various wards. While the Councillors provide political Direction of the District, the executive arm of the District is headed by a Council Chairperson who is assisted by directors covering various functions of the council. The Council Chairperson is directly responsible for the execution of Council development strategies and also provides technical and professional guidance to the Council on matters relating to District development. The Council however, endorses what is to be implemented based on resolutions from its full Council meetings.

The traditional leadership in the area is headed by Chief Sikoongo. Through the traditional lineage and ascendancy guidelines, the chief is flanked by his traditional councils. These are appointed by the Chief and can be dissolved by the Chief as well. These councils head a grouping of villages., who in turn are headed by Village headmen. These headmen are also subordinated by village headman. The Village headmen and Council leaders are close advisers to the Chief.

5.13 Agro-Ecological Zone

Chirundu is in Agro-ecological Zone I, constitutes about 12 percent of the country and covers Luangwa and Zambezi river valleys. It has low altitude and low erratic rainfall with a short growing season. Although it is hot and humid with poor soils in some places, it is suited for growing drought resistant crops. Goat rearing and fishing potential are high as is cattle rearing and cashew nut production in west of Zambezi River in Western Province. Across the river in Chiyaba are many farming activities with crops such as wheat, bananas and animal rearing being predominant.

5.14 Archaeological, Cultural and Historical resources.

The UN Foundation (2004) argue that Heritage Sites carry out numerous functions that are beneficial and essential to human welfare. The nearest cultural and heriatage site is the Chirundu Fossil Forest which is about 21Km off Chirundu-Lusaka road.

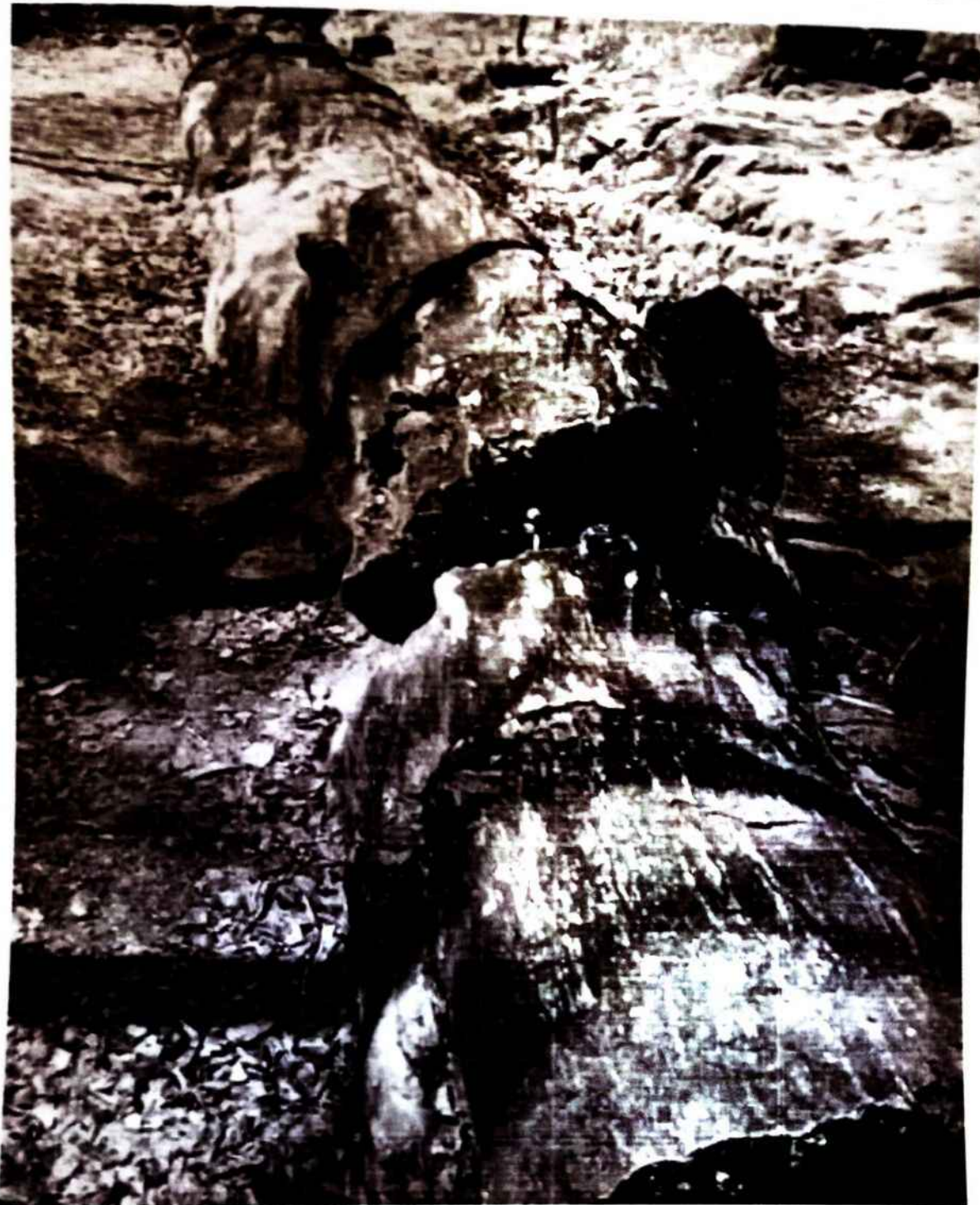


Figure 6: fossil rock in fossil forest

5.15 Built Environment

The main development closest to the project site is the Chiyaba Bridge that separates Chirundu from Chiyaba. About two hundred meters away downstream on the other side of the river is a Zambeef pumping station. Upstream of the proposed site about 1,6 Km is Gwabi lodge where boats park. Another lodge in the nearest vicinity is Kandende safari resort which is in the 600m radius. There is a police post about 300 meters from the bridge towards Chiyaba.

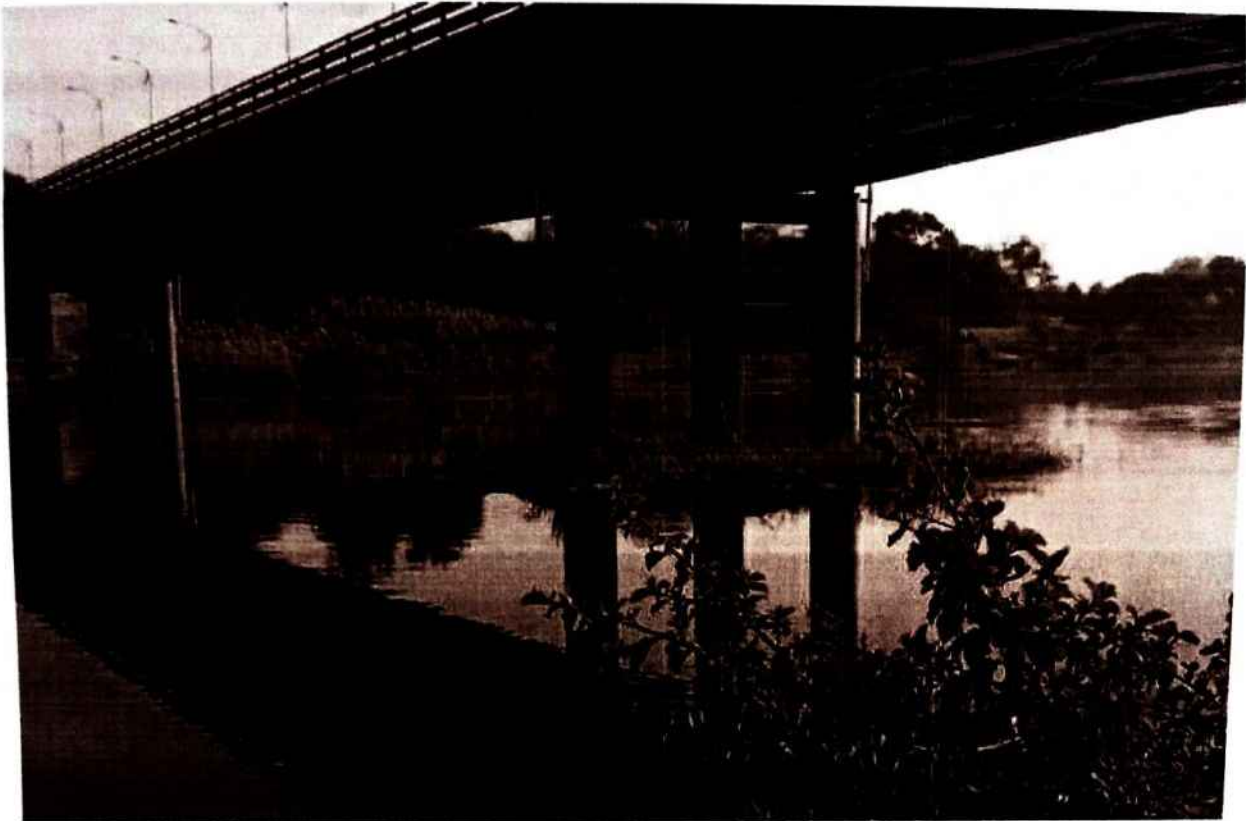
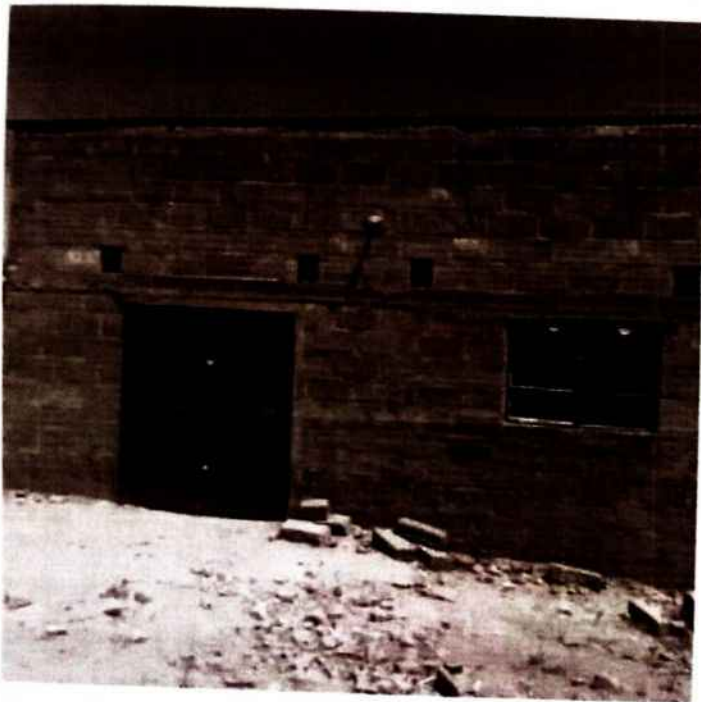


Figure 7: main land mark near the proposed site

The proposed site is on lease and already has developments such as an outside toilet, two small houses which are currently used as storage facilities and one big house which is being by workers working on the aquaculture project.



Figure 8: main house at the project site where workers are staying



6 ENVIRONMENTAL IMPACTS

Positive Impacts

The benefits of this development will either be direct or indirect depending on the service and products to be encountered.

6.1 Socio-economic environment

The positive Impacts of the development of the aquaculture project on the socio-economic environment will include among others; -

- a. The creation of employment opportunity for both skilled and non-skilled labour especially for the local community during construction through to operational phase,
- b. Skills transfer to the local community,
- c. Boosting of local businesses due to the increased buying power
- d. Introduction of new technology

Negative Impacts

Below is a list of potential negative environmental impacts.

6.2 Socio-economic environment

The negative Impacts of the development of the project on the socio-economic environment will include among others;

- a. Impact on public health: Increasing health risks such as HIV/AIDS, Covid-19 etc;
- b. Impact on public health: Introducing of nuisances, such as pests, noise, invasive species and related multiplication of breeding sites.
- c. Human-Animal conflicts arising from interaction with big game on the river such as hippo's and Crocodiles. This conflict will include access to common resource as domesticated animals also use the river for drinking water.

6.3 Physical environment

The negative impacts of the project on the Physical environment will be categorised as follows;

- a. Impact on aesthetics: Changing or altering the natural aesthetic and landscape of the area,
- b. Impact on water quality: Contamination of water due to oil and fuel spillage and sewage waste, dust emissions and fish waste, siltation, poor solid waste management
- c. Impact on Air quality: dust emissions, dead fish smells, vehicular emissions
- d. Impact on Soil erosion: soil erosion from construction activities, oil contamination, poor waste management
- e. Impact on water circulation
- f. Solid waste generation

project

- g. Soil erosion and siltation from land preparation works and disturbance of natural flows of water

Biological environment

The Project will negatively impact the Biological environment as follows; -

- a. Spills and waste will reduce water quality and cause land degradation
- b. Loss of natural habitat from vegetation clearance and noise
- c. Dust and vehicular movements will lead to poor air quality
- d. Disturbance of soils will lead to erosion and siltation
- e. Disturbance of drainages might lead flooding, siltation and land degradation.

7 METHODOLOGY OF IMPACT EVALUATION

In the evaluation of impacts for significance, the procedure used is the liontiev matrix method for impact assessment which is also developed for the South African Mining Industry and is being adopted in Zambia with minor modifications to suit the Zambian situation in respect of Zambian Industries. The descriptions are detailed below and ranked according to impacts, process and evaluation criteria.

Table 5: Ranking criteria

C O N S E Q U E N C E	Nature/ Intensity/ Severity of Impact		
	Low	Impacts affect the environment in such a way that natural, cultural and /or social functions and processes are not affected.	1
	Medium	Impacts affect the environment in such a way that natural, cultural and / or social functions and processes are altered but are reversible.	3
	High	Impacts affect the environment in such a way that natural, cultural and / or social functions and processes will temporary or permanently cease.	5
	Spatial extent of Impact		
	On-site	Impact effect is on project site	1
	Local within the project area	Impact effect can extend to other project sites within or near the project area	2
	Regional within Chirundu and Chiawa	Impact effect can extend to other project sites within Chirundu and Chiawa area	3
	National within Zambia	Impact effect can extend nationally	4
	International	Impact effect can extend beyond the country's borders like Zimbabwean fish cages	5
	Duration of Impact		

project

	Short term	Through dilution either naturally or otherwise and dispersion the impact reduces to insignificant within a short time e.g. one week.	1
	Medium term	Though dilution naturally or otherwise and dispersion the impact reduces to insignificant within the life of the project.	2
	Long term	The impact will cease after the operational life of the project either because of natural processes or by human intervention.	3
	Permanent	Where mitigation either by natural process or by human intervention will not occur in such a way or in such time span that the impact can be considered transient.	4
	Probability of potential occurrence of the impact		
L I K E L Y H O O D	Improbable	The possibility of the impact occurring is very low either because of design or historic experience.	1
	Probable	There is a distinct possibility that the impact will occur	2
	Highly Probable	It is most likely that the impact will occur	3
	Definite	The impact will occur regardless of any preventive measures	4
	Frequency of potential occurrence of the Impact		
	Annually or less	Impact occurs at least once in a year or less frequently	1
	6 months	Impact occurs at least once in 6 months	2
	Monthly	Impact occurs at least once a month	3
	Weekly	Impact occurs at least once a week	4
	Daily	Impact occurs daily	5

N.B. for positive impacts, this changes to appropriate level of beneficial effect.

project

Table 6: Significance Matrix

		CONSEQUENCE												
LIKELIHOOD		3	4	5	6	7	8	9	10	11	12	13	14	
	2	5	6	7	8	9	10	11	12	13	14	15	16	
	3	6	7	8	9	10	11	12	13	14	15	16	17	
	4	7	8	9	10	11	12	13	14	15	16	17	18	
	5	8	9	10	11	12	13	14	15	16	17	18	19	
	6	9	10	11	12	13	14	15	16	17	18	19	20	
	7	10	11	12	13	14	15	16	17	18	19	20	21	
	8	11	12	13	14	15	16	17	18	19	20	21	22	
	9	12	13	14	15	16	17	18	19	20	21	22	23	

Significance is evaluated based on the score of an activity by using Tables 6 and 7 above. Using the totals in Table 9 (columns 6 and 9) for each impact, its overall significance is read out in Table 6. The ranking of the overall significance is as provided in Table 8 below. Description given in Table 7 above are for negative impacts. For positive impacts the wording will generally be the opposite to that of a negative impact. The score is for without mitigation measures. Mitigation measures proposed are aimed at reducing the significance in case of negative impacts or enhancing or maintain the significance in case of positive impacts.

Table 7: classification of overall significance

Level of Significance	Description	Score Range
Low	Where it will not have a significant influence on the environment. Management measures can be proposed to ensure that significance does not increase	5 – 11
Medium	Where it could have a significant influence on the environment unless it is mitigated or managed.	12 – 17
High	Where it would have a significant influence on the environment regardless of any possible mitigation and hence it must be either avoided or managed.	18-23

project

Impact type or Activity	Consequence				Likelihood			Rank score	Impact significance without mitigation	Comment
	N/I /	Spatial Extent	Duration.	Total Score	Prob.	Frequency.	Total Score			
Access to common water resource	3	1	1	5	1	1	3	8	Low negative	The Local community and domesticated animals usually pass through the project site to access the river
Noise pollution.	1	1	1	3	1	5	6	9	Low negative	During construction phase minimal noise impact will be produced
Hazardous waste	1	1	2	4	1	5	6	10	Low negative	Only hazardous waste possible is sanitary waste and contami

project

Impact type or Activity	Consequence				Likelihood			Rank score	Impact significance without mitigation	Comment
	N/I /	Spatial Extent	Duration.	Total Score	Prob.	Frequency.	Total Score			
										nated soils
Influx of people to the area putting a strain on Health (HIV/ AIDS)	3	1	2	6	1	5	6	12	Medium negative	The project will attract a number of workers and their dependants. Further, the increased cashflow in the area will attract in-migrations.
Flooding and Siltation	3	2	3	8	2	2	4	12	Medium Negative	Disturbance of soils and natural drainages can lead to flooding and siltation

project

Impact type or Activity	Consequence				Likelihood			Rank score	Impact significance without mitigation	Comment
	N/I /	Spatial Extent	Duration.	Total Score	Prob.	Frequency.	Total Score			
Soil Erosion	1	1	1	3	2	4	6	9	Low Negative	Vegetation clearing and soil disturbance will lead to soil erosion
Dust emissions and improper waste management	1	1	2	4	2	5	7	11	Low negative	Dust emissions will most likely be experienced during the construction phase though deliveries will also have the potential but minimal
Land degradation and loss of pristine riverine	1	1	3	4	2	4	6	10	Low negative	There will be clearing of the pristine riverine.

project

Impact type or Activity	Consequence				Likelihood			Rank score	Impact significance without mitigation	Comment
	N/I /	Spatial Extent	Duration.	Total Score	Prob.	Frequency.	Total Score			
Soil and water pollution	1	2	2	5	2	1	3	8	Low Negative	This can arise from spillages, leakages and poor waste management
Human -Animal conflicts	1	2	3	5	2	2	4	9	Low negative	Due to proximity to the Kafue river and Chiawa area
Poaching	1	2	3	5	2	2	4	9	Low negative	The presence of foreign national might attract the trade in game meat by local poachers
Conflict as a result	1	2	2	5	2	1	3	8	Low negative	Presence of foreign

project

Impact type or Activity	Consequence			Likelihood			Rank score	Impact significance without mitigation	Comment
	N/I /	Spatial Extent	Duration.	Total Score	Prob.	Frequency.	Total Score		
of cultural differences									national sometimes lead to disputes due to cultural differences
L									

project

8 ENVIRONMENTAL AND SOCIAL MANAGEMENT PLAN

The objective of the Environmental Management Plan (EMP) is to mitigate the possible impacts that may arise that could be foreseen during the implementation phase and therefore maintain the existing environmental quality. The negative impacts are identified as;

- Water quality Impact
- Land degradation
- Loss of access to common property
- HIV/AIDS, Covid-19 risks
- Occupational health and safety
- Human-animal/human conflicts
- Soil pollution
- Siltation and erosion
- Noise pollution
- Potential flooding

The proposed management and mitigation and enhancement measures for the environmental and social commitments are summarized as follows;

Water quality impacts

- All waste such as food stuff packaging materials and feed bags to be disposed of in accordance with local regulations
- Ensuring only road worthy machinery are used to avoid leakages and spillages that can lead to contamination of soils consequently leading to water pollution through storm water.
- Siltation will be managed through silt traps and preservation of the natural pristine riverine.

Access to common property

The following measures will be put in place to avoid conflict

- Ensuring that the fencing of the property does not cover the banks or hinder people from accessing the Kafue river

HIV/AIDS/Covid-19 Impacts

- Awareness and sensitization programs to be facilitated
- Safety measures such as provisions of condoms, counselling and promotion of a stigma free environment for HIV/AIDS. For covid-19, all applicable directive issued by the government will be adhered to.

Waste Management

- Provision of bins
- Subscribing to the local authority waste collection program
- A waste management policy will be available to its employees and enforced
- No open-air burning of waste will be allowed

project

Human – Human/Animal Conflict

Some of these points will also be valid under occupational health and safety

- No unsupervised works access to the River for untrained personnel
- Proper signage for caution against crocodiles to be erected
- Developer to adhere to the labour laws of Zambia
- Ensuring sensitization and awareness program to bridge cultural differences that might lead to disputes
- Ensuring that when it comes to unskilled labour, priority is given to the local people

Fire

- The developer will ensure that fire fighting equipment is available, serviced and that the workers are trained in the operation of the equipment
- All necessary permits from the council will be obtained.
- Proper signage will be set up

8.1 Environmental Monitoring Plan

An overall responsibility for Environmental Management for the proposed project will be the project manager. He will also be responsible for the project monitoring to ensure full compliance to safety, health and environmental standards as pertaining to protected areas regulations.

Table 8: Environmental Management Plan

Aspect	Impact	Mitigation Measure	Frequency of Monitoring	Time Frame	Performance Indicator	Monitoring and Reporting	Cost (ZWK)
CONSTRUCTION AND OPERATION PHASES							
Vegetation to clearing the site prepare the site for construction.	• Loss of biodiversity • Climate change • Land degradation • Loss of water quality	• Limiting clearing to areas marked for construction	Daily	Short term	• Vegetation presence in areas without construction.	Manager	5000
Spills and leakages from machinery on site	Soil and water contamination.	• Ensuing only road worthy machinery is used • Ensuring that all spills are cleaned up and disposed of according to local regulation • Ensuring that a spill lit is present at all times.	weekly	Long term	• Water and soil quality results • No signs of spills on site.	Manager	10,000

Aspect	Impact	Mitigation Measure	Frequency of Monitoring	Time Frame	Performance Indicator	Monitoring and Reporting	Cost (ZWK)
Conflicts between Locals and developer as well as human – animal conflicts	Detering project development	- All leaking machinery to be repaired and serviced. - Provision of proper security - Proper warning signs for caution against crocodiles - Priority of unskilled labour to be given to the local people. - Adherence to local labour laws	Weekly	Long term	- Number of complaints - Number of accidents	Manager	20000
Disturbance of natural drainages	- Flooding - Land degradation - Poor water quality	Installation of culverts	Monthly	Long term	Total Suspended Solids (TSS) levels in water as a result of	Manager	40000

Aspect	Impact	Mitigation Measure	Frequency of Monitoring	Time Frame	Performance Indicator	Monitoring and Reporting	Cost (ZWK)
					- siltation and erosion - Presence of new gullies		
Siltation as a result of land disturbance	- Land degradation - Poor water quality - Drying of the river	- Installation of silt traps - Compacting levelled area - Limiting land disturbance to areas of construction	Monthly	Short term	- TSS levels - Reducing river width and depth	Manager	10000
Poor workmanship, improper management of workshop, negligence and human error	- Loss of property and Injury and loss of life - Breakdown of relations	- Proper implementation of occupational health and safety protocols - Provision of Personal Protective Equipment - Training in safety - Ensuring only qualified people	Monthly	Long term	Number of accidents	Manager	5000

Aspect	Impact	Mitigation Measure	Frequency of Monitoring	Time Frame	Performance Indicator	Monitoring and Reporting	Cost (ZWK)
Noise from construction and operation activities	- Loss of hearing on workers - Public nuisance	- Ensuring that all machinery is properly services - Provision of PPE - Activities to be limited to day time only	Daily	Long term	Number of complaints	Manager	10,000
Dust generation and fumes emissions from machinery as well as foul smells from improper management of waste.	Poor air quality	- Dust suppression through water sprinkling - Proper compaction of disturbed soils - Proper waste management - Maintenance of septic tanks	weekly	Long term	- No foul smells - Good air quality results - Number of complaints	Manager	10000

Aspect	Impact	Mitigation Measure	Frequency of Monitoring	Time Frame	Performance Indicator	Monitoring and Reporting	Cost (ZWK)
Waste generation	• Soil, air and water contamination • Public nuisance	• Proper servicing of machinery • Provision of bins • Subscribing to local authority waste collection program • Ensuring no indiscriminate waste disposal • Optimizing waste reduction and reusing	Weekly	Long term	• Number of complaints • Presence of waste on site and surrounding areas	Weekly	5000
Leakages and spills of hazardous waste from septic tanks	• Water and land contamination	• Septic tanks will be located at a minimum distance of 50m from the river • Proper construction of septic tanks	Bi-annually	Long term	• Foul smells • Faecal coliform levels in water • Disease outbreaks	Monthly	5000

project

8.2 Environmental Monitoring

The identified mitigation measures, proposed monitoring system and environmental audit plan that shall be conducted within 12 months of project implementation sets forth a framework to ensure that all the major issues outlined are addressed in order to minimize adverse impacts during all the three phases of project implementation. Monitoring being the continuous or periodic review of project implementation by project management will be used to assess the Site preparation, construction and operational phases of the project Site construction and implementation of the mitigation measures, the identification of difficulties, the assertion of problem areas, and the recommendation of remedial actions. The purpose of monitoring the development of the project will be to ensure the efficient and effective implementation of the project in order to provide timely information at every stage of the whole process during the three phases of project implementation. Monitoring will be primarily concerned with the delivery process, ensuring that inputs, through activities, are transformed into outputs, and to analyse their quantity and quality.

Environmental monitoring ensures that the impacts have been accurately predicted and that appropriate mitigation measures are being implemented as planned and that they have the expected effects. Identification of potential environmental impacts associated with the Site project indicates a need to design and implement a specific environmental monitoring plan. The monitoring process begins with supervision of implementation. Most of the activities may take place during the implementation stage. The environmental objectives of these activities are to ensure mitigation measures outlined in the contracts are being properly implemented, that environmental contractual measures are being respected, the Site development is going in accordance with the agreed design standards and that no unforeseen negative impacts are occurring as a result of project execution.

The key components of the proposed environmental monitoring plan are presented in the Monitoring Activities and Indicators. While it is appropriate to indicate that progressive environmental monitoring will be practised, this aspect of environmental management is not always possible as some areas only become available as the project progresses. The monitoring plan will be reviewed and adjusted in accordance with project environmental management requirements outlined in this report. Environmental monitoring includes: -

- a) Compliance monitoring and
- b) Impact monitoring

project

9 DECOMMISSIONING AND CLOSURE PLAN

This phase will involve activities aimed at assuaging the adverse effects consequential of the project. Yong Wang Investment Group of Company Limited will ensure that at the end of the project, activities such as removal of the equipment as part of a decommissioning and closure plan that will be submitted to the **ZEMA** for approval.

The objectives of the above is as follows: -

- a) To promote alternative economic activities in the area that are sustainable in the future;
- b) Return the site to conditions capable of supporting life and other tourism related activities;
- c) Prevent potential significant adverse effects on adjacent water resources

In the event of closure, an independent environmental consultant will be engaged to monitor this phase. It is important to note that the land under discussion is on lease and that the infrastructure on site will still be valuable to the land owner hence the structures will be handed over to the land owner.

Project

CONCLUSION

The project, though located in a sensitive environment is expected to have minimal negative environmental impacts on the bio-physical environment. The impacts expected in the operation phases are minimal and will easily be mitigated using the proposed measures.

The project will be implemented in line with the Environmental Management Plan and the monitoring will be done throughout the project life cycle. The project will adhere to all local regulations and project implementation will be done in line with all the mitigation and enhancement measures for maximum realization of the project benefits by both the local community, government and Yong Wang Investment Group of Company Limited.

Project

10 DECLARATION OF AUTHENTICITY OF REPORT CONTENTS

Yong Wang Investment Group of Company Limited hereby affirms and declares that the contents of this report are authentic to the best of their knowledge. No parts of this document shall be reproduced for any reason other than what it is intended for, supporting decision making process for competent authorities on Environmental Management.



.....
Yong Wang Investment Group of Company Limited

MANAGING DIRECTOR

project

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12.1 Annex I: Consent letter



**Sikoongo Main & Lusitu
Sub-Court
P.O. Box 30
Chirundu**

Chen Ref

28th August 2023.

Beatrice Karen Kajembwe
HRH Chieftainess Sikoongo
CC Traditional Council

Wards: *Schizonychia*, *Flutrantho*, *Hydro-antonyama*, *lygantho-lyde*, *Larini*, *Mitrona*, *Noronychia*, *Kapadathia*, *Karonychia*, *Mitronychia*, *Mitronychia*.

VI

12.2 Annex ii: Certificate of incorporation

CP45

(Regulation 46)

Companies Registration No. 120200000304

Serial No. 1127690



Republic Of Zambia

The Companies Act, 2017

(Act No. 10 of 2017)

The Companies (Prescribed Forms) Regulations, 2018

(Section 14)

CERTIFICATE OF INCORPORATION COMPANY LIMITED BY SHARES

This is to certify that YONG WANG INVESTMENT GROUP OF COMPANY LIMITED
is on and from the 14th day of January 2020 incorporated as a COMPANY LIMITED BY SHARES.

Given under my hand and seal at Lusaka, Zambia, this 14th day of January 2020.

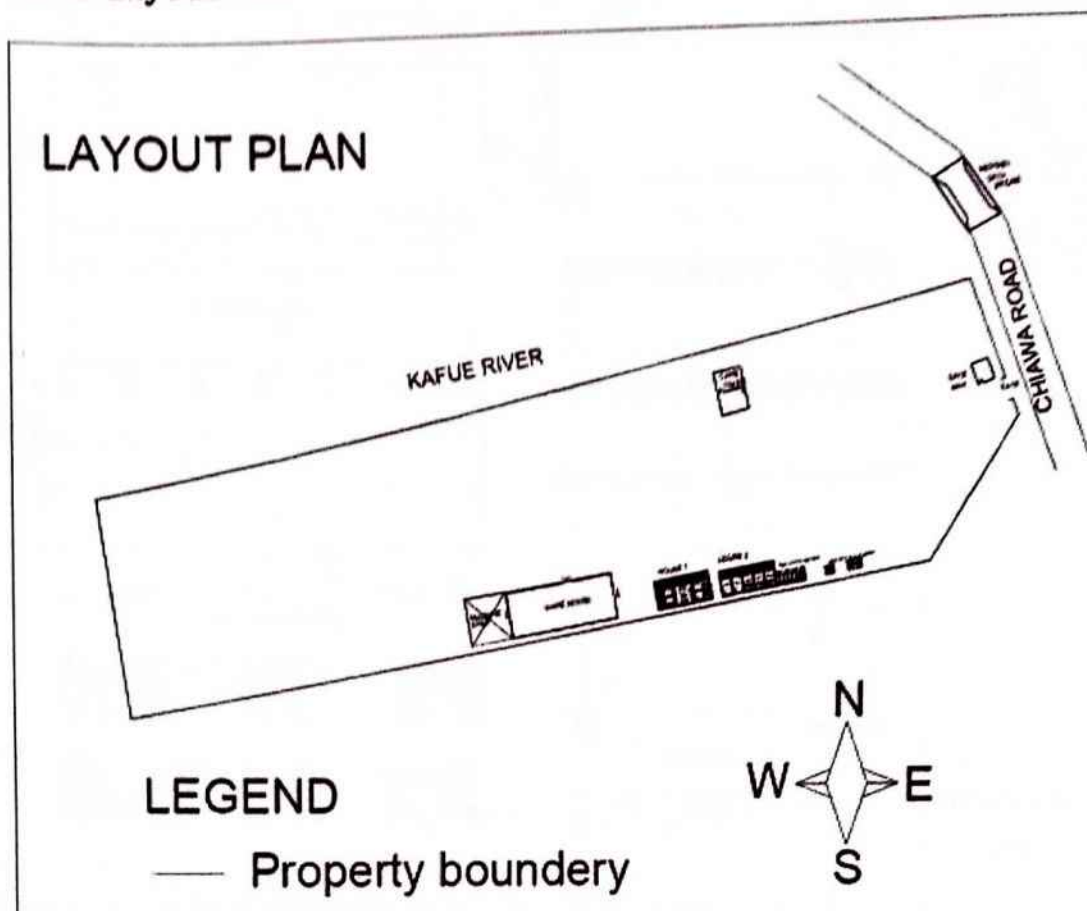


E. M. Ng'andu
Assistant Registrar of Companies

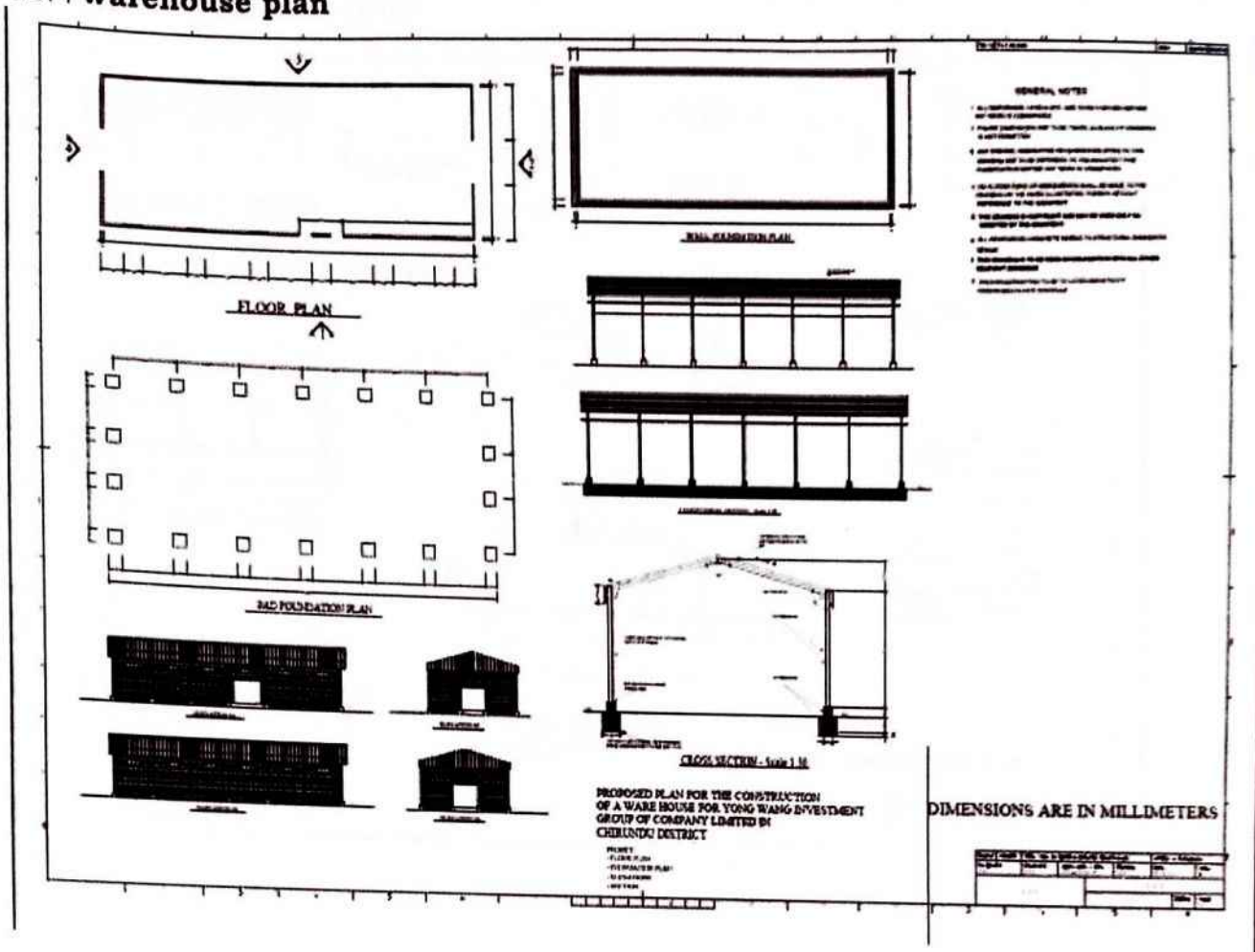
For further details relating to this business visit
<http://www.pacra.org.zm>

project

12.3 Site Layout Plan

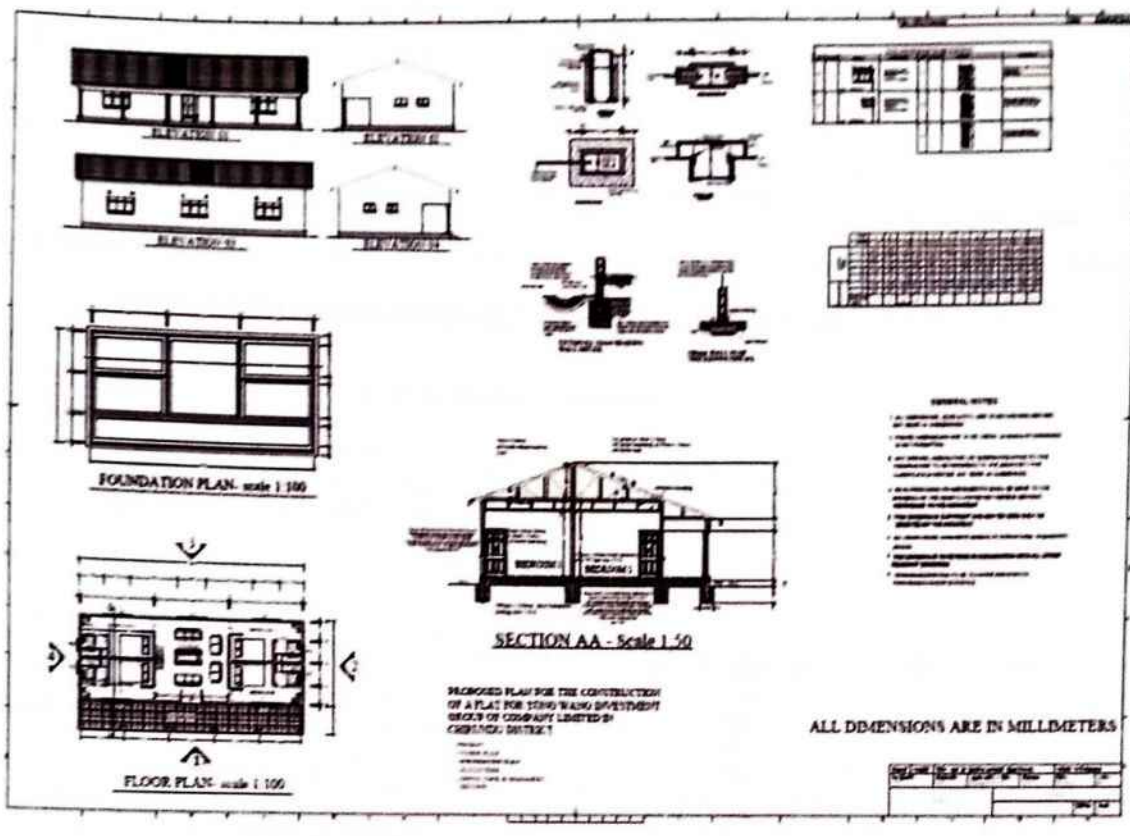


12.4 Warehouse plan



project

12.5 Workers quarters plan



Project

12.6 Water quality results

SCHOOL OF ENGINEERING
CIVIL ENGINEERING DEPARTMENT
ENVIRONMENTAL ENGINEERING LABORATORY

P.O Box 32379, Lusaka

PHYSICAL/CHEMICAL AND BACTERIOLOGICAL EXAMINATION OF WATER

Attn : Yong Wang Group of Companies

Sampled by : Client

Receipt date : 04.09.2023

Report date : 09.09.2023

Laboratory Results

Parameter	Water Sample	ZEMA Standrds	ZABS Standards
pH	7.15	6.0-9.0	6.5-8.0
Conductivity ($\mu\text{S}/\text{cm}$)	252	4,300	1,500
Total suspended solids (mg/l)	<1.0	100	-
Total Dissolved Solids (mg/l)	126	3,000	1,000
Turbidity (NTU)	1.15	15	5.0
Biochemical Oxygen Demand (as mg O_2/l)	0.8	50	-
Chemical Oxygen Demand (as mg O_2/l)	<3.2	90	-
Sulphates (mg/l)	<0.0001	1,500	400
Nitrates (as $\text{NO}_3\text{-N}$ mg/l)	<0.0001	50	10.0
Bacteriological Results			
Feecal coliforms (#/100ml)	75	5,000	0

Tests carried out in conformity with "Standard Methods for the Examination of water and Wastewater APHA, 1998".

Tested by: D. Mkandawire
Chief Engineer

Checked & Approved by: D. Mkandawire
Chief Engineer

